BIO-ENERGY PRODUCTION SYSTEM

Standard Training Manual & Protocols

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Understanding the Energy Loop

A step-by-step technical guide to turning raw waste manure into off-grid heat and electricity.

Step 1: Raw Waste Collection

Gravity Slurry Flow

Waste slurry drops from animal pens through elevated floor slats, flowing naturally along concrete gutters poured at a **2% downward slope**.

Quality Control Filter

The operator must physically inspect and remove heavy dirt, sticks, rocks, or plastic debris to prevent critical clogs inside piping manifolds.

Step 2: Slurry Conditioning

The 90:10 Ratio

Dilute collected solid waste with water to achieve a balanced **90% water to 10% solid** ratio, ensuring optimum fluid flow.

Warm Dilution

Ensure dilution water is above **15°C**. Shocking the digester with freezing water will stall active microbial cultures.

Paddle Agitation

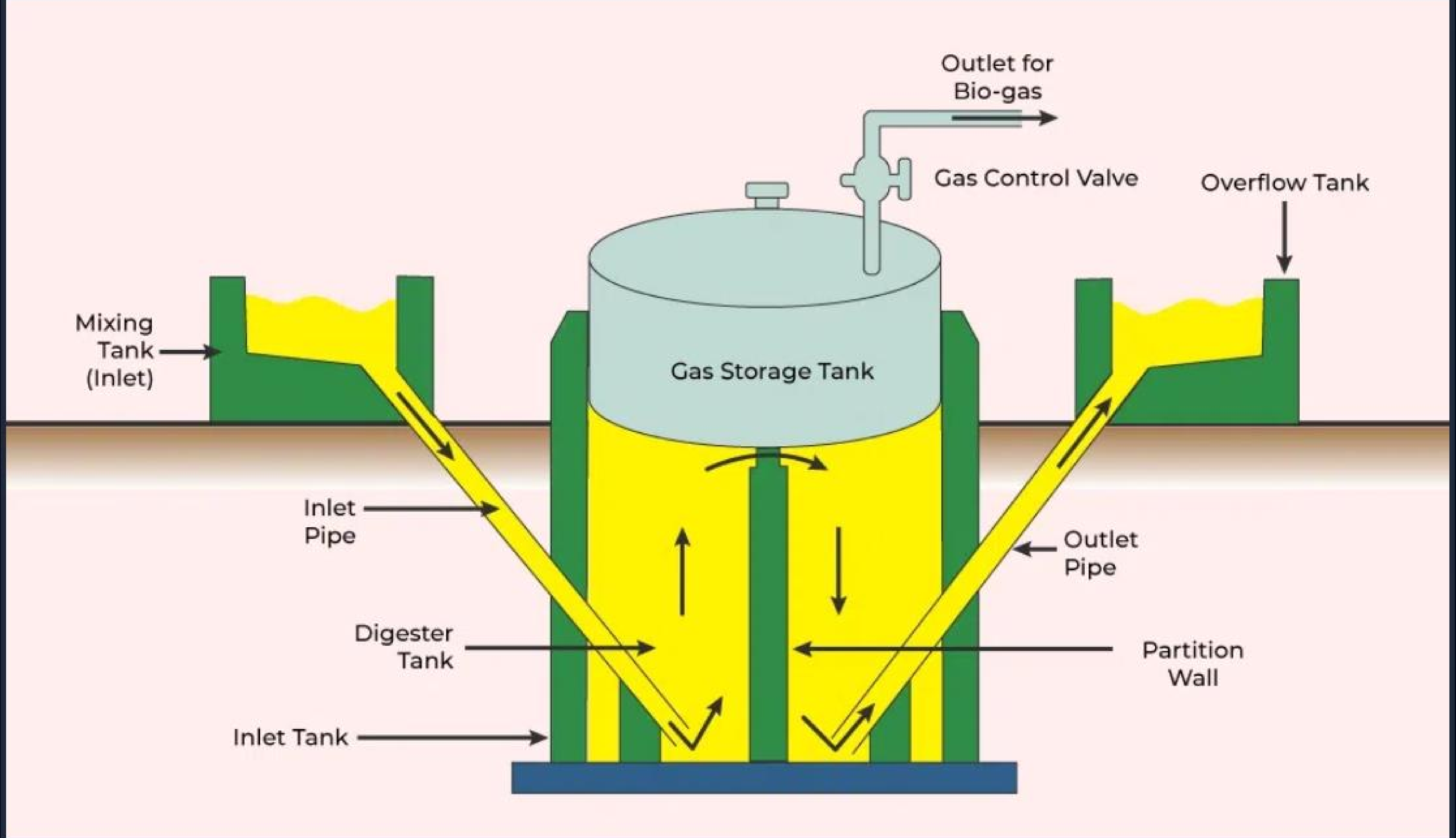
Stir the mixture for 5 minutes inside the intake sump to break up clumps, achieving a completely smooth fluid texture.

Step 3: Inside the Digester Tank

Anaerobic Decomposition

Conditioned slurry is siphoned into the sealed airtight digester. Friendly microbes digest the organic waste, releasing raw methane gas.

- **No Oxygen Permitted:** Oxygen ingress ruins methane microbial growth entirely.
- **Acid-to-Gas Balances:** Microbes convert complex acids into clean methane.
- **Continuous Output:** Gas rises to the dome, and heavy bio-slurry settles down.



Step 4: Gas Scrubbing & Drying

- **Hydrogen Sulfide (H_2S) Scrubbing:** Raw gas passes through inline iron-filing filters to neutralize highly corrosive sulfur elements that damage internal engine surfaces.
- **Moisture Trapping (Drying):** Warm gas flows through cold condenser tubes, allowing water vapor to collect and drain, delivering dry biogas to the generator.
- **Low-Pressure Bag Storage:** Scrubbed, dry biogas is routed to heavy-duty rubber storage bags, stabilized at a safe system pressure of **20 mbar**.

Step 5: Fueling the 12kVA GenSet

Mechanical Methane Fueling

Clean biogas enters the engine manifolds of our modified 12kVA generator set to generate clean electricity.

- **Combustion Cycle:** Air and biogas are balanced to ensure efficient engine ignition.
- **Continuous Power:** The generator converts combustion forces into electric current.

BESS Synchronization

Electricity is routed back to charge the 30kWh BESS system at night, providing stable energy independent of public utility networks.

Step 6: Radiant Sub-Floor Heating

Exhaust Capture

Hot engine exhaust gases are routed through a stainless steel water-jacket heat exchanger, creating hot steam.

Manifold Routing

Hot steam travels through insulated sub-floor pipes located beneath the elevated plastic nursery floors.

Constant 32°C Base

Sub-floor heat keeps nursery floors at a warm, comfortable 32°C without drawing electric utility power.

Unified System Flowchart

1. Waste Sump

Swine manure and solar water blend to achieve a 90:10 fluid ratio.

2. Digester Tank

Anaerobic microbes digest slurry, producing raw methane.

3. Scrubbing Unit

H_2S sulfur and moisture are removed from the gas loop.

4. Heat & Power

12kVA GenSet generates BESS electric volts and sub-floor steam.

Daily Maintenance Schedule

System Component	Action Protocol Checklist	Standard Frequency
Slurry Sump	Measure slurry pH levels to maintain a neutral 7.0 pH.	Every Shift Start
H_2S Filter	Swap iron filings when gas tests show sulfur residues.	Every 90 Days
Moisture Trap	Drain accumulated water from gas loop condensation valves.	Every Friday
Exhaust Valve	Test thermal bypass steam valves for smooth mechanical action.	Weekly

Action Plan: System Recovery

Troubleshooting System Faults

pH <

Action Plan: System Recovery

If pH drops below 6.2, excess acid has killed methane-producing microbes, causing gas output to crash entirely.

- **Step 1:** Stop feeding raw manure slurry into the

Approved for Operator Site Training

PRESTIGE COSMIC VENTURE

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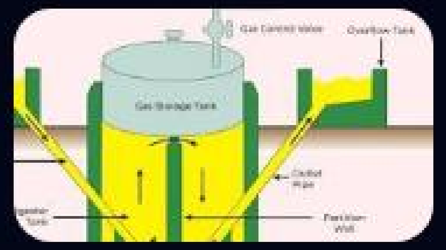
Image Sources



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